

# One-Step Specular Highlight Removal with Adapted Diffusion Models

Mahir Atmış<sup>1</sup> Levent Karacan<sup>2</sup> Mehmet Sarıgül<sup>1</sup>  
<sup>1</sup>Çukurova University <sup>2</sup>Gaziantep University

## Problem & Motivation

**Goal:** Removing specular highlights from the surfaces of the objects.

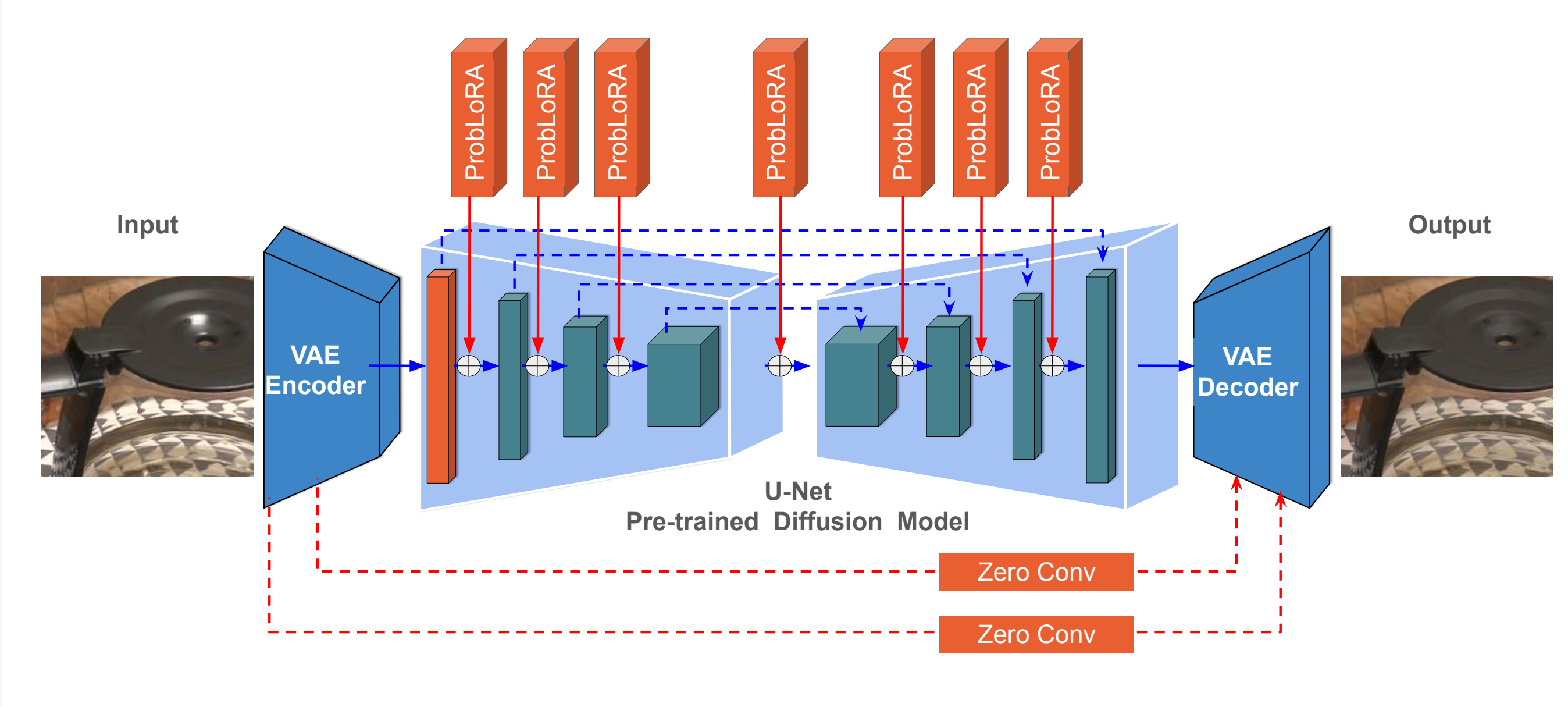
**Challenge:** Loyalty to the original surface color and texture

**Opportunity:** Utilizing Stable Diffusion Turbo (SD Turbo) with a new adaptation mechanism, improved efficiency and adaptability.

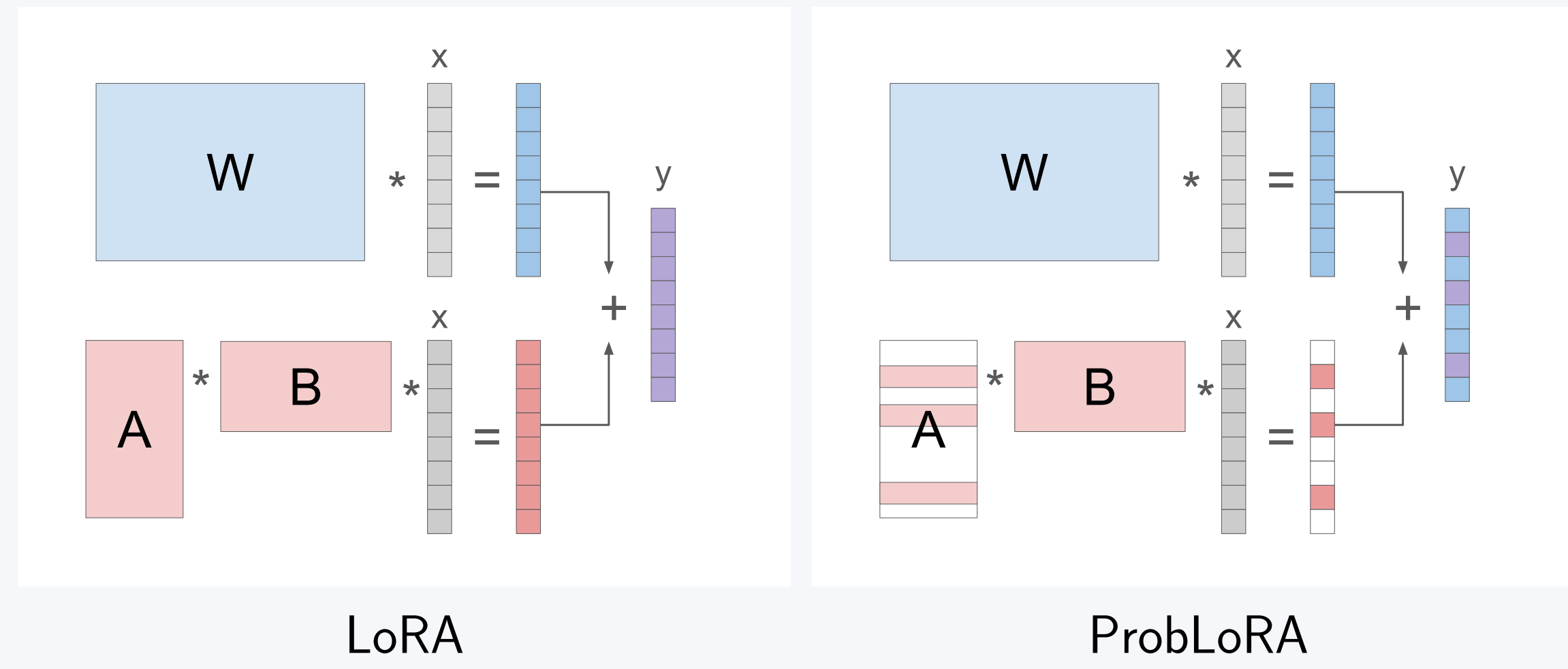
## Key Contributions

1. One-step diffusion-based model for specular highlight removal
2. ProbLoRA, a novel modification of LoRA designed to adapt the diffusion model for specular highlight removal efficiently
3. State-of-the-art performance on specular highlight removal tasks

## Architecture



Overall Architecture



LoRA

ProbLoRA

## Method Overview

$$y = Wx + ABx \quad (1)$$

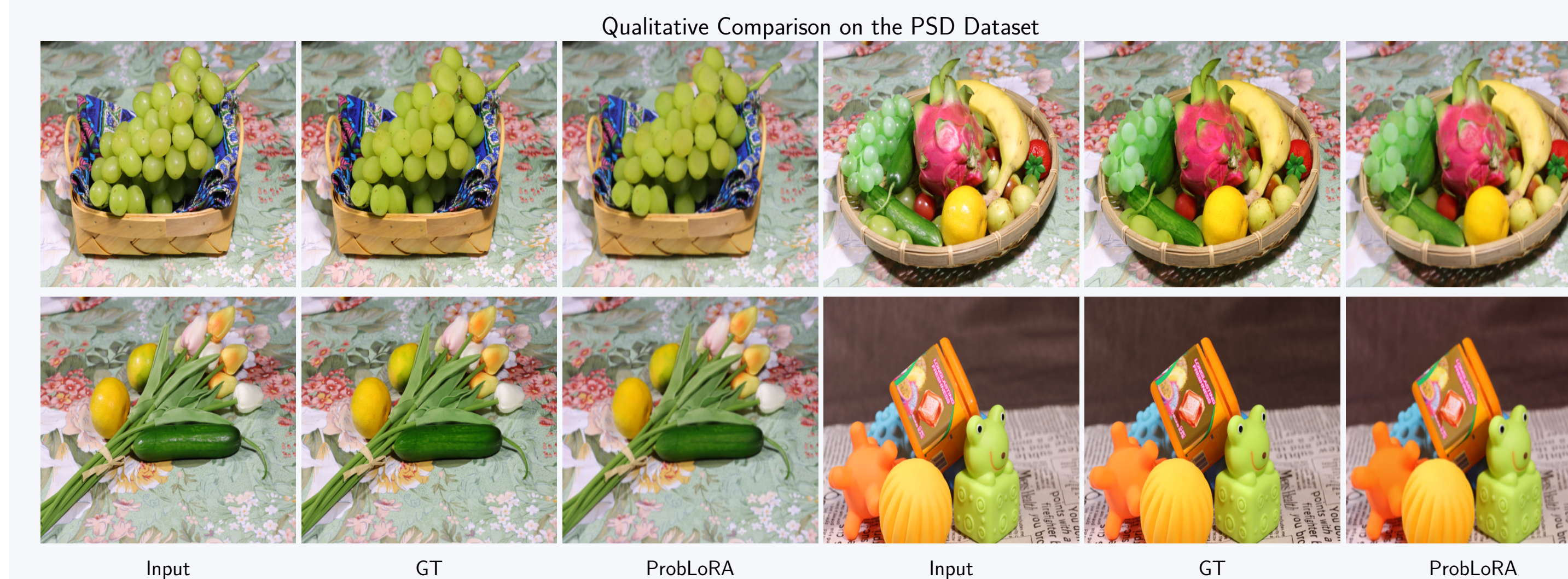
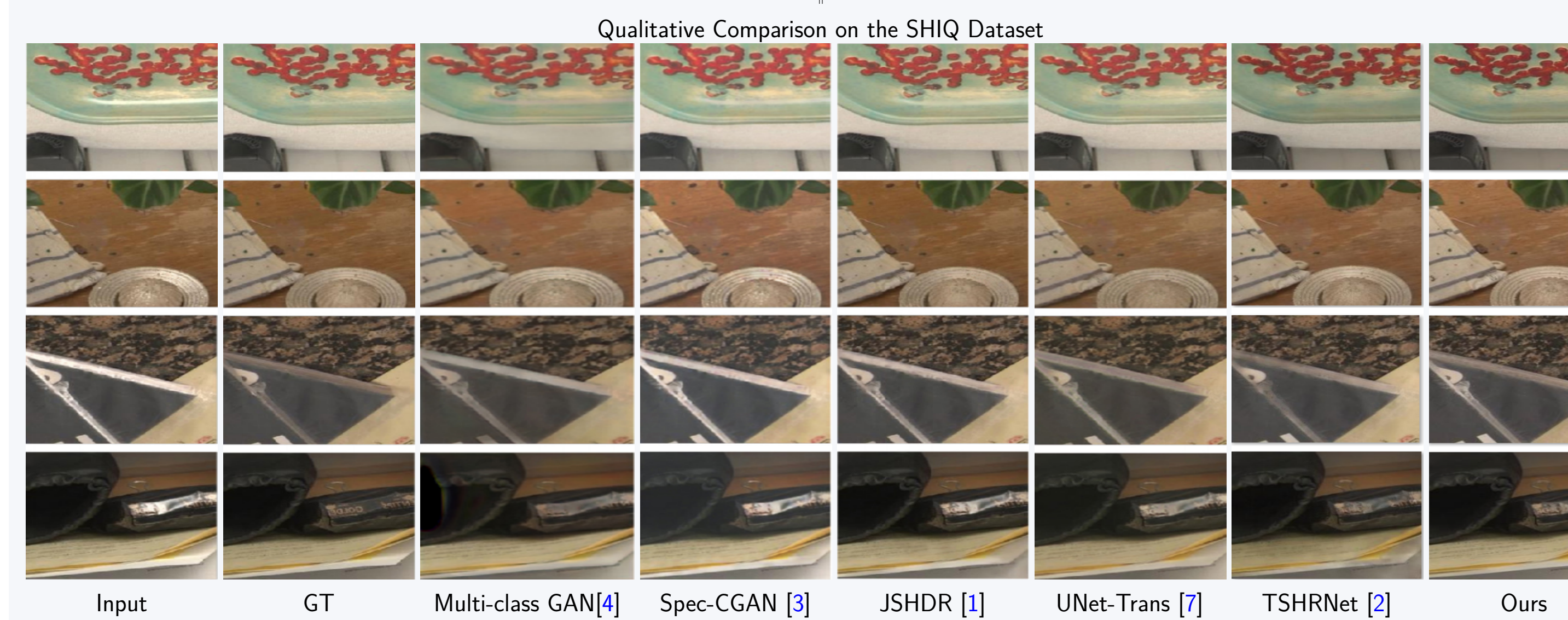
$$y_i = W_i x + p(a_i) a_i Bx \quad (2)$$

$$p(a_i) = \frac{1}{1 + \exp\left(-\frac{M(a_i)}{\tau}\right)} \quad (3)$$

$$M(a_i) = \max_{1 \leq k \leq r} \left\{ \frac{a_{ik}}{\sqrt{r}} \right\} - \frac{1}{r} \sum_{k=1}^r \frac{a_{ik}}{\sqrt{r}} \quad (4)$$

where  $A \in \mathbb{R}^{d_{out} \times r}$  and  $B \in \mathbb{R}^{r \times d_{in}}$  are low-rank matrices,  $r \ll \min(d_{in}, d_{out})$  is rank,  $a_i$  represents the  $i$ -th row vector ( $i = 1, 2, \dots, d_{out}$ ) of  $A$ ,  $p(a_i)$  is the probability of selecting row  $a_i$ ,  $M(a_i)$  is a max-mean measurement inspired by KL-divergence-based sparse attention mechanisms [9], which quantifies the importance of each row with temperature parameter  $0.5 \leq \tau \leq 1$  to control the sharpness of the probability

## Qualitative Results



## Experimental Results

### Quantitative Comparison on SHIQ, SSHR, and PSD Datasets

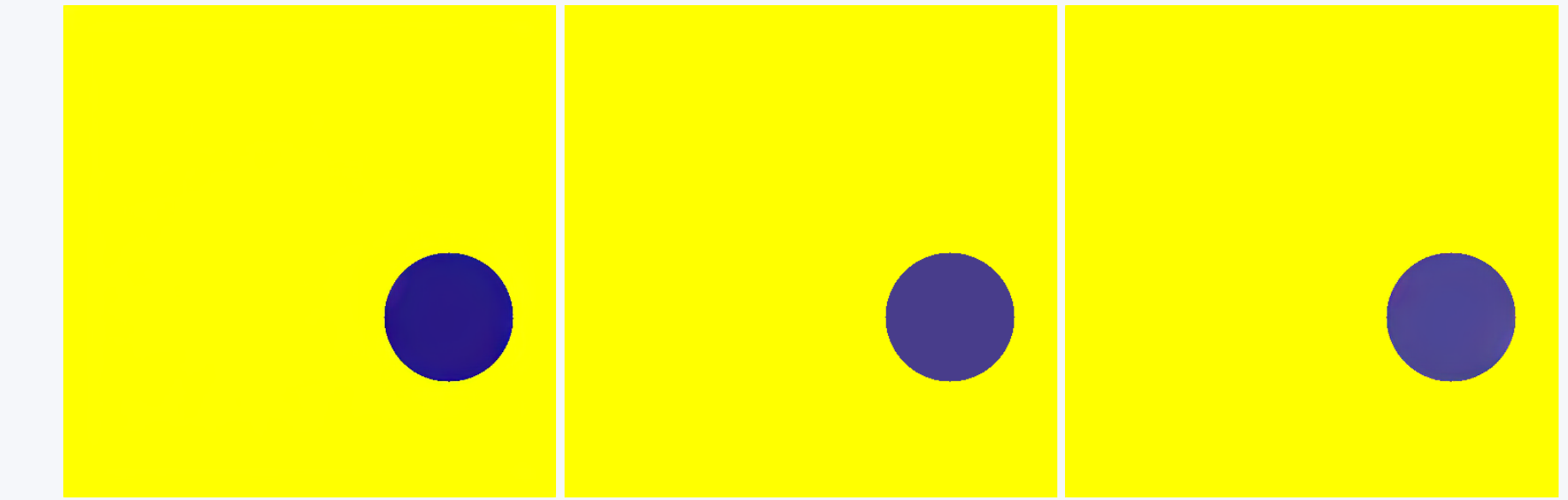
| Dataset     | SHIQ            |                 | SSHR            |                 | PSD             |                 |
|-------------|-----------------|-----------------|-----------------|-----------------|-----------------|-----------------|
|             | PSNR $\uparrow$ | SSIM $\uparrow$ | PSNR $\uparrow$ | SSIM $\uparrow$ | PSNR $\uparrow$ | SSIM $\uparrow$ |
| Fu [1]      | 34.13           | 0.860           | 26.98           | 0.895           | 21.52           | 0.883           |
| Wu [8]      | 23.42           | 0.920           | 25.73           | 0.894           | 21.80           | 0.880           |
| TSHRNet [2] | 25.58           | 0.933           | 28.90           | 0.940           | 22.76           | 0.903           |
| Pan [5]     | 34.34           | 0.966           | -               | -               | -               | -               |
| Baseline    | 33.94           | 0.966           | 34.12           | 0.957           | 29.23           | 0.949           |
| Ours        | <b>36.19</b>    | <b>0.971</b>    | <b>35.76</b>    | <b>0.965</b>    | <b>29.99</b>    | <b>0.957</b>    |

### Ablation on Temperature Parameter ( $\tau$ ) — SHIQ Dataset

| Dataset               | SHIQ            |                 |                  |
|-----------------------|-----------------|-----------------|------------------|
| Metric                | PSNR $\uparrow$ | SSIM $\uparrow$ | MSE $\downarrow$ |
| Temperature( $\tau$ ) |                 |                 |                  |
| 0.5                   | 33.664          | 0.964           | 0.085            |
| 0.6                   | 34.268          | 0.966           | 0.085            |
| <b>0.7</b>            | <b>36.190</b>   | <b>0.971</b>    | <b>0.059</b>     |
| 0.8                   | 36.079          | 0.971           | 0.059            |
| 1.0                   | 33.275          | 0.965           | 0.086            |

## Generalization

We evaluate ProbLoRA on the Fill50K image generation task, where it achieves a significantly lower FID score (6.788) compared to LoRA (10.384) [6], indicating improved perceptual quality and better alignment with the target distribution as can also be seen below. These results highlight ProbLoRA's robustness across architectures and its adaptability to diverse generative tasks, including free-form image completion.



LoRA

GT

ProbLoRA

## Future Work

Extend **ProbLoRA** to Large Language Models (LLMs) to evaluate its scalability and generalization capability.

Apply ProbLoRA to **Image-to-Image Translation** tasks (e.g., Night2Day, Monet2Photo) and compare performance in terms of *speed*, *accuracy*, and *memory efficiency*.

Explore integration with other diffusion-based architectures to analyze whether probabilistic adaptation improves stability and convergence.

## Acknowledgment

This work was supported by the Scientific and Technological Research Council of Türkiye (TUBITAK) under project number 123E671.

## Links & Contact

🔗 Code: <https://github.com/mahiratmis/ProbLoRA>  
 🧑 Demo: [Hugging Face Space](#)  
 ✉ Email: [matmis@cu.edu.tr](mailto:matmis@cu.edu.tr)  
 🌐 Website: [https://mahiratmis.github.io/prob\\_lora/](https://mahiratmis.github.io/prob_lora/)



Scan for project website

## References

- Gang Fu, Qing Zhang, Lei Zhu, Ping Li, and Chunxia Xiao. A multi-task network for joint specular highlight detection and removal. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 7752–7761, 2021.
- Gang Fu, Qing Zhang, Lei Zhu, Chunxia Xiao, and Ping Li. Towards high-quality specular highlight removal by leveraging large-scale synthetic data. In *Proceedings of the IEEE/CVF International Conference on Computer Vision*, pages 12857–12865, 2023.
- Isabel Funke, Sebastian Bodenstedt, Carina Riediger, Jürgen Weitz, and Stefanie Speidel. Generative adversarial networks for specular highlight removal in endoscopic images. In Baowei Fei and Robert J. Webster, editors, *Medical Imaging 2018: Image-Guided Procedures, Robotic Interventions, and Modeling*, volume 10576 of *Society of Photo-Optical Instrumentation Engineers (SPIE) Conference Series*, page 1057604, March 2018.
- John Lin, Mohamed El Amine Seddik, Mohamed Tamaazousti, Youssef Tamaazousti, and Adrien Bartoli. Deep multi-class adversarial specular removal, 2019.
- Lu Pan, Hongwei Zhao, and Runze" Wu. Towards specular highlight removal through diffusion model. In *Pattern Recognition and Computer Vision*, pages 497–509. Springer Nature Singapore, 2025.
- Gaurav Parmar, Taesung Park, Srinivasa Narasimhan, and Jun-Yan Zhu. One-step image translation with text-to-image models. *arXiv preprint arXiv:2403.12036*, 2024.
- Zhongqi Wu, Jianwei Guo, Chuanqing Zhuang, Jun Xiao, Dong-Ming Yan, and Xiaopeng Zhang. Joint specular highlight detection and removal in single images via unet-transformer. *Computational Visual Media*, 9(1):141–154, 2023.
- Zhongqi Wu, Chuanqing Zhuang, Jian Shi, Jianwei Guo, Jun Xiao, Xiaopeng Zhang, and Dong-Ming Yan. Single-image specular highlight removal via real-world dataset construction. *IEEE Transactions on Multimedia*, 24:3782–3793, 2021.
- Haoyi Zhou, Shanghang Zhang, Jieqi Peng, Shuai Zhang, Jianxin Li, Hui Xiong, and Wancai Zhang. Informer: Beyond efficient transformer for long sequence time-series forecasting. In *Proceedings of the AAAI conference on artificial intelligence*, volume 35, pages 11106–11115, 2021.